

Jayanth Mummalaneni

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Summary

Results-driven Industrial & Quality Engineer with hands-on expertise in continuous improvement, lean manufacturing, and quality systems. Proven track record of optimizing production efficiency, reducing waste, and driving cost-saving initiatives through tools like Six Sigma, PFMEA, VSM, and Root Cause Analysis. Experienced in coordinating cross-functional teams, managing supplier quality, and leading APQP/PPAP documentation. Seeking to leverage process optimization skills to drive operational excellence in manufacturing environments.

Strengths

Continuous Improvement: Lean Manufacturing, PDCA, Kaizen, VSM, 5S, Root Cause Analysis, Bottleneck Elimination

Quality Systems: APQP, PPAP, FMEA, MSA, SPC, CAPA, ISO 9001, IATF 16949

Industrial Engineering: Time Studies, Setup Reduction, BOM Management, Cost Reduction, Process Mapping

Tools & Software: MINITAB, AutoCAD, Microsoft Excel (Advanced), PowerPoint, Word, Outlook

Certifications: Six Sigma, Quality Tools 8-D, DMAIC, TQM, SPC, PFMEA

Experience

Pioneer Plastics – Warren, MI

July 2024 – Present

Quality Engineer

- Reduced customer complaints by 50% by standardizing label visibility through BOM updates and IQMS integration.
- Cut scrap rate from 10% to 4.17% on cupholder liners by troubleshooting resin dryer settings, ducts, and hoppers, saving \$400/month.
- Delivered \$7,000 in cost savings on speaker grill defects by updating DFMEA, implementing Gauge R&R, and ensuring rework met EOL inspection requirements.
- Lowered nonconformance tickets from 16% to 3% by retraining operators/QCs and restructuring workstations, generating \$1,000 in monthly savings.
- Eliminated all assembly-related NCTs by moderating operator work pace and cross-checking procedures, saving \$300/shipment.
- Prevented 250 lbs of raw material waste by implementing barcode tracking for poly pro contamination and aligning BOMs in IQMS.
- Lowered incorrect assembly setup scrap rate from 50% to 10% by retraining operators on machine inputs and maintaining tool holders.
- Saved 2 days of machine and operator costs by resolving resin color mismatch and implementing master setup and raw material tracking.
- Reduced lead times by 15% by leading PPAP process improvements for 60+ parts in collaboration with suppliers and internal teams.
- Cut defect rates by 10% by driving PFMEA analysis and implementing corrective actions using 5S, VSM, and root cause analysis.
- Improved job setup times by 10% and reduced recurring defects by 20% through streamlined production launches and updated job books.
- Handled APQP metrics, improved on-time supplier delivery, and resolved COPQ issues across 100+ supplier interactions.
- Managed CAPAs and NCRs across multiple teams, ensuring quality issue accountability and readiness for audits.
- Created and maintained BOMs and IQMS documentation, improving accuracy in procurement and cost control.

D.V. Polymers India Pvt Ltd – Hyderabad, India

August 2021 – June 2022

Production Quality Engineer

- Increased workflow efficiency by 12% through error analysis and continuous improvement initiatives using MINITAB.
- Achieved 100% customer acceptance by conducting dimensional analysis and Gage R&R studies across multiple processes.
- Reduced procurement costs by 5% by optimizing ultra-grade raw material usage through life-cycle and cost-performance analysis.
- Delivered projects 2 days ahead of schedule by eliminating non-value-added steps using Root Cause and Pareto Analysis.
- Ensured documentation accuracy for over 30 technical drawings, enhancing compliance and supporting smooth project execution.

Raam Group – Hyderabad, India

April 2021 – July 2021

Automotive Technician Intern

- Conducted inspections and tire/brake testing on 500+ vehicles, improving safety and diagnostic accuracy.
- Reduced diagnostic time and improved customer satisfaction by mastering Xentry DAS software on 100+ vehicles.

- Performed 50+ post-repair test drives, validating performance and identifying residual issues in alignment with QA standards.
- Supported senior technicians in component installations (differentials, control units, catalytic converters), increasing installation efficiency and process reliability.

Key Projects

Quality Control in Injection Molding – Real-Time Lean Tool Application

- Reduced injection molding inefficiencies by 15% through value stream mapping and capability analysis.
- Presented findings at IEOM Bangladesh 2022; Won 3rd place for process optimization research.

Capacity Loss Mitigation – Lean Tools Optimization

- Increased value-added time by 20% by redesigning workflows using VSM and Root Cause Analysis.
- Applied PDCA and bottleneck analysis to optimize batch production and reduce pharmaceutical manufacturing lead times.

Education

Lawrence Technological University – M.S. in Industrial Engineering

Michigan, USA

- Focus: Lean Six Sigma, Production Planning & Control, Manufacturing Systems

Sreenidhi Institute of Science & Technology — B.S. in Mechanical Engineering

Hyderabad, India

- Focus: Manufacturing | Coursework: Thermodynamics, Metrology, Manufacturing Processes

Activities

Lawrence Technological University – Part-Time Student Assistant

Michigan, USA

- Streamlined inventory control systems, cutting material overhead by 80% and improving resource visibility.
- Maintained 100% confidentiality in managing faculty and student records.
- Led setup of 100+ academic/executive meetings and coordinated research tasks, enhancing operational efficiency.
- Recognized for attention to detail and creating a high-performance, organized environment.